

	Internal Exam	Assignment	Project Evaluation	End Semester Examinations
CO 1	✓			✓
CO 2	✓			✓
CO 3	✓	✓		✓
CO 4		✓		✓
CO 5		✓		✓
CO 6			✓	✓

Programme	BSc Statistics				
Course Code	ACT3MN202 (P)				
Course Title	Life Contingencies				
Type of Course	Minor				
Semester	III				
Academic Level	200 - 299				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	Basic knowledge of data, variables, charts and graphs, Basic computer skills				
Course Summary	The aim of this course is to provide a grounding in mathematical and statistical modelling techniques that are of particular relevance to actuarial work, including survival models and their application.				

Course Outcomes (CO):

CO	CO Statement	Cognitive	Knowledge	Evaluation
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		Level*	Category#	Tools used
CO1	Calculate gross premium of assurance and annuity contracts using gross future loss random variable	Ap	P	Instructor-created exams / Quiz
CO2	Describe and calculate reserve under assurance and annuity contracts using prospective and retrospective method of valuation.	AP	P	Instructor-created exams / Home Assignments
CO3	Memorize the basic concepts of mortality profit.	R	C	Seminar Presentation / Group Tutorial Work
CO4	Define and calculate death strain at risk, expected death strain and actual death strain	An	P	Instructor-created exams / Home Assignments
CO5	Understand the basics of Multiple decrement model	U	F	One Minute Reflection Writing assignments
CO6	Analyse gross premium and reserve with sample data with spread sheet	Ap	P	Viva-Voce/Practical Assignment
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hours (45 +30)	Marks (70)
I	Gross premiums		11	20
	1	Gross premiums, Gross future loss random variable, Principle of equivalence	3	
	2	Determining gross premiums using the equivalence principle	2	
	3	Annual premium contracts	2	
	4	Premiums payable m times per year	2	
	5	Calculating gross premiums using simple criteria other than the equivalence principle	2	
	Sections from References: Unit 1: 20.1, 20.2 &20.3 [Ref 1] Unit 2: 20.3.2 [Ref 1] Unit 3: 20.3.5 [Ref 1]			

	Unit 4: 20.3.7 [Ref 1] Unit 5: 20.4 [Ref 1]			
II	Gross premium reserves		13	20
	6	Gross premium reserves, Why hold reserves	2	
	7	Prospective reserves, Calculating gross premium prospective reserves,	3	
	8	Calculating prospective reserves that satisfy probabilities	2	
	9	Retrospective reserves, Retrospective accumulations, Gross premium retrospective reserve	2	
	10	Equality of prospective and retrospective reserves	2	
	11	Net premium reserves for conventional without profit contracts.	2	
	Sections from References: Unit 6: 21.0 & 21.1 [Ref 1] Unit 7: 21.2 & 21.2.1 [Ref 1] Unit 8: 21.2.2 [Ref 1] Unit 9: 21.3, 21.3.1, 21.3.2 [Ref 1] Unit 10: 21.4 [Ref 1] Unit 11: 21.6 [Ref 1]			
III	Mortality profit		12	15
	12	Mortality profit on a single policy	2	
	13	Mortality profit on a portfolio of policies	2	
	14	Allowing for death benefits payable immediately	2	
	15	Allowing for survival benefits annuities	2	
	16	Allowing for different premium or annuity payment frequencies	1	
	17	Calculation of mortality profit for policies involving two lives.	3	
	Sections from References: Unit 12: 24.1 (24.1.1 to 24.1.4) [Ref 1] Unit 13: 24.2 [Ref 1] Unit 14: 24.3 [Ref 1] Unit 15: 24.4 [Ref 1] Unit 16: 24.5 [Ref 1] Unit 17: 24.6 [Ref 1]			
IV	Competing risks		9	15
	18	Health insurance contracts	1	
	19	Multiple state models	2	
	20	Valuing continuous cash-flows using multiple	2	

		state models		
	21	Designing the multiple state model	2	
	22	Multiple decrement models, Multiple decrement tables	2	
	Sections from References: Unit 18: 25.1 [Ref 1] Unit 19: 25.2 [Ref 1] Unit 20:25.2.2[Ref 1] Unit 21: 25.2.3 [Ref 1] Unit 22: 25.3&25.4[Ref 1]			
V	PRACTICUM		30	
	Do practice problems in spreadsheet from any 5 units of the given list and one additional problem decided by the teacher-in-charge, related to the content of the course. Other units listed here may be used as demonstrations of the concepts taught in the course.			
	1	Gross future loss random variable		
	2	Gross premium- Whole life		
	3	Gross premium- endowment		
	4	Annual premium contracts		
	5	Gross premium reserve		
	6	Prospective and retrospective reserve		
	7	Mortality profit		
	8	Multiple decrement tables		
Books and References: 8. Institute of Actuaries Act Ed. Study materials CM1. 9. Neill, Alistair, Heinemann, (1977): <i>Life contingencies</i> . 10. Bowers, Newton Let al (1997): Actuarial mathematics, society of Actuaries, 2nd Ed 11. Jones, H.E & Long, D.L (2005): Principles of Insurance: Life, Health and annuities. LOMA 12. Dickson, D.C.M; Hardy M.R; Waters, H.R.-Actuarial Mathematics for life contingent risks: 2nd ed. Cambridge University Press (2013) 13. Neill, Alistair, Heinemann, (1977): <i>Life contingencies</i> . 14. Shailaja R. Deshmukh- Actuarial Statistics-an introduction using R, 3 rd Ed.				

Mapping of COs with PSOs and POs:

	PSO 1	PSO 2	PSO 3	PSO4	PSO 5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6
CO 1	-	2	2	2	-	-	2	1	2	-	-	-
CO 2	3	3	-	2	1	-	3	1	2	2	-	-
CO 3	1	2	-	2	-	-	2	-	2	-	-	-

CO 4	3	2	2	1	-	-	3	-	2	3	-	-
CO 5	3	2	-	-	-	-	3	-	2	-	-	-
CO 6	1	2	2	-	3	3	2	2	1	-	2	1

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Midterm Exam
- Programming Assignments (20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics:

	Internal Exam	Assignment	Project Evaluation	End Semester Examinations
CO 1	✓			✓
CO 2	✓			✓
CO 3	✓	✓		✓
CO 4		✓		✓
CO 5		✓		✓
CO 6			✓	✓

FOUNDATION COURSES IN STATISTICS

SYLLABUS